properties of, 21t, 28t
spine quantum number of, 22
strong interaction force and, 18
weak interaction force and, 18
LET, *see* Linear energy transfer
Leukaemia, 463f, 463–464
Level of mother decay rate, 387
Light, 344–346, 345f
kinetic energy of, 19
stellar, 347, 347t
Light water reactors (LWRs), 599f
conversion ratios and fuel utilization in, 620f
coolant for, 598–599
fuel reprocessing from, 704f, 709
LOCA in, 636
plutonium formation in, 699–700, 701t
prevalence of, 598, 598t
reaction products
balance of, 618f
build-up in, 617f
shim control in, 634–635
Light years, 340
Linear accelerators, multiple-stage, 499f, 501f, 502–505, 502f
Linear density, of target material, 523
Linear energy transfer (LET)
iron oxidation and, 224, 224f
radiation values, 172t
in water reactions, 221
Linear momentum, conservation of, 87–88
Liquid hydrogen bubble chamber, reaction in, 15, 16f
Liquid metal cooled fast breeder reactor (LMFBR), 674
plutonium formation in, 699–700, 701t
Liquid partition chromatography (LPC), of radioactive tracers, 555, 564–566, 565f, 566f
Liquid scintillator detectors, 272–273, 272t, 273f
Liquid volumes, radioactive tracers used in determination of, 587–588
Liquid-drop model, 322
of fission, 333–336, 334f, 335f
semiempirical mass equation, 333–334, 335f
shell corrections for, 335–336, 335f
Liquid–liquid extraction, *see* Solvent extraction
Lithium (Li)
isotope separation for, 55–56
isotopic data for, 34t
isotopic ratio of, 41
Lithium-drifted detectors, 266–267, 267f, 268f
LLWs, *see* Low-level wastes
LMFBR, *see* Liquid metal cooled fast breeder reactor
Load factor, 631–632
LOCA, *see* Loss of coolant accident
Logarithmic distribution coefficient, 551, 552f
London Dumping Convention, 402
Lorenz equation, 22–23
Loss of coolant accident (LOCA), 636
Low radiation doses, 446
somatic effects of, 469–471
epidemiological results, 470f, 471–473
study problems for, 472f, 473–474
Low-intensity measurements, 283
Low-level wastes (LLWs), treatment and deposition of, 721f, 728
LPC, *see* Liquid partition chromatography
Lr, *see* Lawrencium
Lung cancer, 463–464, 481–482
LWRs, *see* Light water reactors

M

m, *see* Metastable
M, *see* Atomic weights
Macroscopic absorption cross-section, 182–183
Macroscopic cross-section, 315–316
Magnetic confinement, 641, 641f
Magnetic fields
ions in, 35–36, 35f
in mass spectrometer, 37
Magnetic moment (μ_s), 26, 131f, 132, 134–135
atoms and nuclei, 134–135
nuclear, 135
single particles, 132–134
Magnetic orbital quantum numbers (m_l), 132–133
Magnox reactor (GCR)
prevalence of, 598, 598t
radionuclide release from, 638, 639t
spent fuel from, 695
Mass, *see* Relativistic mass, Rest mass
Mass attenuation coefficient, 182–183
Mass classification, of elementary particles, 20

Mass defect (ΔM_A), 69–72, 71t
  binding energy and, 71t
  of helium, 72
Mass energy, 298–299
Mass excess (δ_A), 70, 72
  binding energy and, 71t
Mass number (A), 32
  atomic number relationship to, 32
  conservation of, 88
  in radioactive decay, 6–7
Mass separator procedures, 541f, 542
Mass spectrometer, 35
  applications of, 37–40
  for chemical analysis, 37
    instrumental, 40
    for reactions, 38
  common type of, 37
  construction of, 35, 35f–36f
  for gas purity analysis, 40
  for isotopic dilution, 38–40
  for isotopic masses and abundances, 35–37
  molecular weight determination with, 38
  for nuclear reaction product measurement, 40
Mass yield curves, for fission
  with protons, 330–331, 330f
  with thermal neutrons, 328, 328f
Mass–energy relation of Einstein, 19
Master–slave manipulators, 488–489, 488f
Matter waves, 20
Maximum credible accident (MCA), 632–633
MCA, *see* Maximum credible accident
MCAs, *see* Multichannel analysers
MCP, *see* Microchannel plate detectors
Md, *see* Mendelevium
Mean free path, 315–316
Mean lethal dose, 457
Measured count rate (R), 112
Measurement techniques, *see* Detection and measurement techniques
Mediators, 29
Medium-level wastes (MLWs), treatment and deposition of, 721f, 728
Mendelevium (Md), 424–427
Mental retardation, 465–466
Mesons
  as antiparticle, 23
  mass classification of, 20
  properties of, 21t
  quarks of, 27
  substructure of, 17
Metals, radiation effects on, 216–217
Metastable (m), 32–33
Microchannel plate detectors (MCP), 274
Microwave radiation, Big Bang and, 349–350
Milky Way galaxy, 340, 341f, 342f
  age of, 367–368
Minor actinides, 423
Mixed radioactive decay, 111–112, 115f
Mixed uranium-plutonium oxide (MOX) fuel elements, 690
Mixer-settler, 791–792, 791f–792f, 795–796
Mixing, radioactive tracers used in, 586
m_j, *see* Total magnetic angular momentum quantum number
m_l, *see* Magnetic orbital quantum numbers
MLWs, *see* Medium-level wastes
Models, 125–126, *see also* Deformed nuclei model, Nuclear model, Single-particle shell model, Standard Model
  atomic, 7
  collective nuclear, 144–145, 145f
  compound nucleus, 307, 309–310
    lifetime of, 323–324, 323f
  direct interaction, 324–327, 325f, 326f, 327f
  of fission, 333–336, 334f, 335f
  liquid-drop, 322
  optical, 321
Moderation, 305
Moderators, 305, 600
Molecular weight determination, with mass spectrometer, 38
Molten metal purification, 715
Molten salt electrorefining, 709
Molten salt extraction, 709
Molten salt reactor (MSR), 670
Molten salt transport, 709
Moment of inertia, 130–131
Monte Carlo calculation, 327, 327f
Mössbauer effect, 196–198, 196f, 197f, 199f
Most probable charge, 329–330
Most probable kinetic energy, 44–45
Most probable velocity, 45
MOX fuel elements, *see* Mixed uranium-plutonium oxide

MSR, *see* Molten salt reactor
Multichannel analysers (MCAs), 278–279
Multiple stage continuous solvent extraction, 791–793, 791f–793f
Multiple-stage linear accelerators, 499f, 501f, 502–505, 502f
Multistage processes, for isotope separation, 52–55, 53f
Muon neutrino
  creation of, 24
  properties of, 25, 28t
Muons
  from cosmic radiation, 377
  creation of, 24, 376–377
  decay of, 24, 376–377
  discovery of, 374
  properties of, 21t
  weak interaction force and, 18
M, *see* Solar mass

N

n, *see* Principal quantum number
N, *see* Neutrons
NAA, *see* Neutron activation analysis
Negatron emission, *see* Electron emission
Neon (Ne)
  fission of, 73–74
  isotopes of, 6
    separation of, 57
Neptunium (Np), 418–419
  isotopes of, 418
  occurrence and production of, 419
  uses of, 419
Neptunium radioelement series, 5f, 386–387
Nernst equation, 552
Net peak area, 282–283, 282f
Neutrinos, 23–26, *see also* Antineutrino
  Big Bang theory and, 25–26
  creation in beta decay, 23–25, 92–94, 93f
  detection of, 23–25
    of solar, 25
    of supernova, 25
  flavours of, 24–25
  properties of, 21t
  rest mass of, 25
  spin of, 92
  weak interaction force and, 18
Neutron activation analysis (NAA), 558–561, 559t, 560f, 562t
Neutron age, 611–612
Neutron capture, 319–320, 320f
  neutron stars, 366–367
  processes, 361–367
  in radionuclide production, 518–523, 519f
  rapid process, 364–366, 365f, 365t
  slow process, 361–362, 365t
  supernova explosions, 359f, 362–364, 364f
Neutron cycle, in nuclear reactors, 607–611, 608f, 610t
Neutron diffraction, 320–321
Neutron emission, 67–68, 86–87
  with fission, 331
Neutron leakage, 611–613
Neutron multiplication factor, 597
Neutron multipliers, 311
Neutron radiography, 204–205, 204t
Neutron reflector, 608
Neutron stars, 343, 366–367
Neutron-deficient isotopes, 514
Neutron-rich isotopes, 514
Neutron-rich nucleus, 67–68
Neutrons (N)
  absorption of, 189
  binding energy of, 128, 129f
  creation of, 350–351
  decay of, 95
  delayed, 329
  detection of
    fission chamber for, 259–260
    proportional counters for, 259
  energy spectrum of, 602–603, 603f
  generators of, 508–510, 509f, 509t
  inorganic substance exposure to, 218
  interaction properties of, 136t
  magnetic moment of, 134
  mass of, 33
  moderation of, 606–607, 606t
  nuclear radiation absorption and, 165t
  in nuclear reactor, 597
  nuclear stability and, 65–67, 66f
    proton ratio to, 67–69, 69f
  nucleus approach of, 128, 129f